

LITTLE BLUE BOOK NO. 720
Edited by E. Haldeman-Julius

The Intelligence of Invertebrate Animals

Maynard Shipley

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INTELLIGENCE OF INVERTEBRATE ANIMALS ***

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Edited by E. Haldeman-Julius

The Intelligence of Invertebrate Animals

Maynard Shipley

HALDEMAN-JULIUS COMPANY

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THE INTELLIGENCE OF INVERTEBRATE ANIMALS

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THE INTELLIGENCE OF INVERTEBRATE ANIMALS

INTRODUCTION

No one doubts today that all Vertebrates, from the Fish to Man, possess more or less intelligence—namely, the power of forming memory associations and of learning by experience. But when we come to the study of Invertebrates (e. g., Insects, Spiders, Mollusks, Worms), we meet with much divergence of opinion among the authorities. Many competent investigators have been led to the conviction that the capacity to learn by experience, to form memory associations, leading to intelligent adjustment of behavior to environment, resides only in those animals which possess a true cerebrum and central nervous system, such as is found only in the Vertebrates. Below these forms, they conclude, lie only “instinctive reactions” to stimuli—“purposeful action without consciousness of purpose” (Hartmann). With increasing complexity of the nervous system, arises the neural mechanism for memory association, the basis of intelligence; and to “inherited habit” (instinct) is added deliberate (intelligent) purposeful adaptation of conduct to a consciously desired end, the underlying motive being avoidance of pain and the attainment of pleasure, or satisfaction.

A simple illustration of the modification of mere instinct by associative memory, plus pleasurable or painful effects, is afforded by Lloyd Morgan’s experiments with chicks.

As is well known, no sooner is a young chick hatched than it begins—by instinct—to peck at all sorts of objects of about a certain size, whether they are edible or not. If, however, one of these objects happens to be a caterpillar of nauseous taste, upon meeting a similar caterpillar a second time the chick appears to remember its first painful experience, and refrains from pecking at it, and may even scrape its bill on the ground as if to wipe off the bad-tasting substance, so vividly is it recalled. This is an example of associative memory and “learning by experience,” avoiding the repetition of acts previously attended by pain or displeasure. “What distinguishes intelligence from instinct is,” says Professor Holmes, “that in the latter the connections between acts are based upon hereditary organization, whereas

in the former they are established through experience.... We have in this modification of instincts through the pleasurable or painful effects they produce the beginning of intelligence.”

Such notable psychologists as Mach, Loeb, Holmes, Thorndyke, Morgan, Whitman, Baldwin, and many others accept as a criterion of intelligence in animals this capacity for memory association, constituting what the metaphysicians call *consciousness*.

For most modern psychologists, *an interruption of consciousness is merely an interruption of the activity of associative memory*. They speak of the extent of associative memory in the animal kingdom instead of the extent of consciousness among animals.^[1]

If an animal possesses the mechanism for associative memory, then it possesses, unquestionably, the mechanism for intelligence. Just what this necessary mechanism is, and in what group of animals it appears in its simplest form, is at present a debatable question—not to say an unsolved problem.

On the positive side, we may assume that if an animal can “learn by experience”—associate the memory of a previous experience with a present situation, and profit by this association—it possesses intelligence.

On the other hand, failure on the part of an experimenter to train an animal to react in a desired way does not afford proof of the absence of intelligence—of associative memory. It may only prove that wrong methods have been employed by the experimenter.

Some authorities have found evidences of intelligent reaction to stimuli, not only in such Invertebrates as the Crab, Crayfish and Horseshoe Crab, but also in the Octopus, Starfish and even among the lowly group familiarly known as “Worms.”

Binet went farther and wrote a learned work on “The Psychic Life of Micro-organisms,” in which considerable intelligence was attributed to Infusorians. But that was thirty years or more ago.

While no competent writer would assert today that psychic life is entirely absent in the Protozoa (one-celled animals), it is now generally conceded

that no adequate proof of intelligent response to stimuli among unicellular animals has so far been forthcoming.^[2]

In the opinion of the late Dr. Jacques Loeb, the claim of a number of investigators that associative memory and therefore of intelligence as here defined, is possessed by even such multicellular forms as Worms, Starfish, Sea Urchins, Actinians, Medusae and Hydroids, is unwarranted by the experimental data. "Claims for the existence of such memory in these groups of animals," remarks Loeb, "have frequently been made, but such claims are either plain romance or due to a confusion of reversible physiological processes with the irreversible phenomena of associative memory. The less a scientist is accustomed to rigid quantitative experiments, the more ready he is to confound the reversible after effects of a stimulus—e. g., the effects due to an increase in hydrogen ion concentration—with indications of associative memory. Learning is only possible where there exists a specific organ of associative memory, the physical mechanism of which is still unknown."

Loeb admitted that associative memory "exists in most mammals," also in birds. In the lower Vertebrates this capacity, he thought, was only *occasionally* found: "Tree-frogs, for example, can be trained, upon hearing a sound, to go to a certain place for food. In other Frogs, *Rana esculenta*, for instance, no reaction is as yet known which proves the existence of associative memory. Some Fishes evidently possess memory; in Sharks, however, its existence is doubtful. With regard to Invertebrates, the question is difficult to determine. The statements of enthusiasts who discover consciousness and resemblance to man on every side should not be too readily accepted."

We must now add, that equal caution should be used in regard to those authors who contend that intelligence does not exist among Invertebrates, but only "instinctive reactions to stimuli." To describe an act as "instinctive," moreover, does not explain much. G. Bohn asks: "What is instinct?" and answers: "A word."

Mr. Garrett P. Serviss recently received a letter from a citizen of Philadelphia from which I quote the following paragraph:

"You speak of the intelligence of the Spider. Understanding and reasoning go hand in hand with intelligence; there is no intelligence without

reasoning or understanding, hence it can be applied to Man. How do you apply it to the Spider? So, as regarding the Bee and the Ant, would you not think that their intelligence is simply 'instinct,' peculiar to their species for self-preservation?"

Mr. Serviss replied (in part) as follows:

"I see no reason for denying intelligence to animals ranking below Man in the scale. Both the lower animals and the human species possess what are called instincts, which are said to act spontaneously, without the aid of reasoning. No animal, indeed, possesses a greater number of instincts than Man.

"But if the distinction between instinct and intelligence is to be based on the employment of conscious reasoning or intelligence, this faculty cannot be denied to the lower animals, because I believe that many of them do exercise a power of reasoning, i. e., of drawing conclusions from observation.

"Natural history books are full of instances of exhibitions of intelligence by Dogs, Cats, Elephants, Horses and many other creatures. It is true that some naturalists insist that all of the apparently intentional and reasoned acts of such creatures are merely manifestations of instinct, or unconscious responses to external stimuli, but all naturalists are not of that opinion."

No, they are not.

Prof. S. J. Holmes says: "Psychologists nowadays with comparatively few exceptions agree in regarding intelligence not as a faculty standing in sharp contrast to instinct, as was formerly taught, but as one resting on a foundation of instinct, and gradually growing out of behavior of the purely instinctive type. The term intelligence is used here in the wider sense as embracing all those forms of profiting by experience through the formation of associations. It therefore includes psychic activity ranging from simple associative memory to complex trains of reasoning."

Paley, in his "Natural Theology," defines instinct as "a propensity prior to experience and independent of instruction"; while Spencer ("Principles of Psychology") states that instinct is a "compound reflex action," which is terse, if not explanatory.

Hartmann's "purposeful action without consciousness of the purpose" is a contradiction in terms. "Purpose" implies consciousness of end in view. One cannot readily believe that when an Orang-utan builds its platform or "nest" in the boughs of a tree in the jungle it does not purpose (*intend*) to rest and sleep on it; or that when a Beaver cuts down a tree in such wise that it invariably falls in the direction of its *need* it does not *know* that it is constructing a hut to live in that will meet the requirements of the situation in a running stream.

Dr. Wm. T. Hornaday, in his authoritative recent work, "The Minds and Manners of Wild Animals," (1922) says:

"Instinct is the knowledge or impulse which animals or men derive from their ancestors by inheritance, and which they obey, *either consciously or subconsciously*^[3] in working out their own preservation, increase and betterment. Instinct often functions as a sixth sense."

Again: "While avoiding the folly of idealism, we also must shun the ways of the narrow mind, and the eyes that refuse to see the truth. Wild animals are not superhuman demigods of wisdom; but neither are they idiots, unable to reason from cause to effect along the simple lines that vitally affect their existence.... Some animals have more intelligence than some men; and some have far better morals."

Nevertheless, it is still "correct" to say: "Man alone possesses reason." Fundamentalists, and even many scientists, find this phrase acceptable. The "Age of Reason" began with the appearance of Man on earth, according to orthodox views. "Instinctive behavior" explains—for many scientists—all acts even of such highly organized creatures as Ants, Bees and Wasps. Below this come "tropisms"—obligatory movements made by the organism as a whole in response to the varied stimuli of its environment.

Then we have the category of "purely reflex" actions of the organism—the definite but unconscious reaction of the organism to stimulation of certain nerve-cells.

In the higher microscopic organisms it is admitted that any one of several reactions to a given stimulus might *occur*, the organism "trying" (unconsciously) one reaction after another, on the hit or miss, or "trial and

error” method. Man, of course, learns by some faculty other than “trial and error”—maybe.

Finally, and lowest in the scale of animal evolution, we come to the Amœba, a unicellular organism apparently lacking even in a rudimentary nervous system. But this microscopic speck of carnivorous protoplasm seeks and *pursues* its zigzagging prey!

While there is no proof that psychic life begins only with pluricellular animals, thus denying any sort of consciousness of purpose to even the highest groups (Ciliata) among unicellular organisms (Protozoa), it must be admitted that no conclusive evidence of the presence of mind in these lowly organisms has yet been presented. I shall therefore turn at once to the Metazoa (many-celled animals) for examples of the intelligence of Invertebrates; beginning with the lowest in which manifestations of mind are said to appear, namely, the Earthworm.

FOOTNOTES:

[1] Cf. Loeb, Jacques, “Comparative Physiology of the Brain and Comparative Psychology,” New York, 1900.

[2] For an excellent discussion of this problem see Holmes, Prof. S. J., “The Evolution of Animal Intelligence,” Pages 63-89, New York, 1911; and Jennings, Prof. H. S., “Behavior of Lower Organisms,” New York, 1906. See also, Day and Bentley, “A Note on Learning in Paramœcium,” *Journal of Animal Behavior*, 1, 67, 1911; and Watkins, G. P., “Psychical Life in Protozoa,” *American Journal of Psychology*, 11, 166, 1900.

[3] Italics mine.

EARTHWORMS

In the preceding introduction it was stated that a certain degree of intelligence had been attributed to animals as low in the scale of evolution as Earthworms. No less an authority than Charles Darwin was responsible for this conclusion.^[4] This great naturalist had observed that these lowly creatures had developed the peculiar habit of plugging up their burrows with dead leaves. He noted that leaves were drawn into the burrows by methods best suited to their particular shape. For example, linden leaves were drawn in by their tips, while no attempt was made to pull in leaves of the rhododendron by this method, these leaves being larger at the tip than at their base. On the other hand, pine needles, which frequently occur in pairs, with a common base, were never seized by the small end, since the separated ends of the needles would cause trouble. They were invariably attacked at the compact basal end, which is smaller than the mouth of the burrow.

“Instinct” may guide Worms in this nice discrimination. But Darwin tested them with materials unknown to them or to their ancestors. He cut up paper into triangular forms and placed it at the disposal of the Worms. To his astonishment the brainless creatures almost invariably seized the bits of paper by the most acute angle in carrying them to their burrows. Darwin therefore concluded that an intelligent choice of means to end had been made. Similar experiments were later made by Hanel, and Darwin’s results were verified; but the apparently intelligent reactions of the Worms were interpreted as “more or less complex reflexes in relation to the form and chemical nature of the objects drawn in.” As to which of these two interpretations is more in harmony with the observations, I leave to the reader’s own judgment.

Generally speaking, it is probably best to accept the guiding principle formulated by Prof. Lloyd Morgan, namely, “In no case may we interpret an action as the outcome of the exercise of a higher psychical faculty, if it can be interpreted as the outcome of one which stands lower in the psychological scale.”^[5]

Darwin's conclusion seems to be justified by some experiments quite recently carried out by Professor Heck of the University of Prague, who gathered together for his purpose some five hundred Worms. The Worms were introduced into a passage shaped like a capital T, carved from a block of wood, and covered with a glass plate so that the movements of the animals might be watched. When they came to the junction, about half of them turned one way and half the other. Then the apparatus was arranged so that those Worms which took the left-hand passage received a mild but probably disagreeable shock. At first they did not seem to know what to make of this; but after they had all been through the experience about 200 times, they nearly all took the right-hand turn. When the electrodes were then moved to the right-hand passage, the Worms learned to shift to the left-hand after only 65 trials—evidently showing something beyond the operation of mere chance.

Commenting on this attitude of Earthworms to learn by experience, a writer in *The Scientific American* (April, 1924) remarks:

“In the human sense, Earthworms have no brains; their nervous systems consist of a series of little ganglions, or nerve centers, on the under side of the Worms and connected with each other by nerve fibers. If the Worms were cut in two, the fragments still showed the ability to distinguish between the safe and the unpleasant road to travel, indicating that the Earthworm remembers in every one of its ganglions, and that it is able to learn and profit by experience.”

In other words, the phenomena of associative memory, hence the capacity to learn by experience, seems to occur not only in animals devoid of cerebral hemispheres—long regarded as the sole seat of intelligence—but in organisms possessing no cerebro-spinal nervous system. In Invertebrates the mechanisms which allow associative memory “will probably be found in the supra-œsophageal ganglion” (Loeb).

“The Earthworm has no specialized sense organs, it has neither eyes to see, nor nose to smell, nor ears to hear with. Still, although it is apparently deaf, it is not devoid of the power of appreciating those stimuli which in us excite the sensation of sight or smell. A strong light suddenly turned on the anterior end of the body will cause the Worm instantaneously to withdraw into its burrow, and Worms readily recognize the presence of such favorite

food as onions and raw meat. Their sense of touch is well developed and they are very sensitive to vibrations; for instance, a stamp of the foot on the ground will cause all those in a certain radius to disappear into their burrows. It is further possible that Earthworms possess other senses with which we are totally unacquainted.”^[6]

It is quite evident, from what has been said above, that remarkable responses to environmental stimuli are made by animals quite devoid of a brain and lacking in those sense organs popularly assumed to be necessary to such responses of the organism as have just been described.

FOOTNOTES:

^[4] See his classical work, “The Formation of Vegetable Mould Through the Action of Worms, with Observations on Their Habits,” New York, 1883.

^[5] Cf. Morgan. Prof. C. L., “Animal Life and Intelligence,” 1891; “Habit and Instinct,” 1896; “Animal Behavior,” 1900.

^[6] Shipley and MacBride, “Zoology,” 4th Ed., 1920.

STARFISH

Memory, according to Romanes, begins with the Echinoderms—e.g., Starfish, Sea-anemones, etc.; and Preyer's extensive experiments with Starfish led him to believe that he had "discovered indubitable indications of intelligent action" in the case of Ophiurus ("brittle star"). The question is still under debate, with the weight of authority on the side of Loeb, Jennings, Gaser and others, whose experiments led them to the opposite conclusion.

Thomson finds that Starfish learn to meet new difficulties in new ways. If they cannot surmount their difficulties one way, they will try another.^[7]

Starfishes are nearly always hungry and they consider Sea-urchins a delicacy. But Sea-urchins are armed with small but sharp "three-bladed screws" on their backs. Knowing this, the Starfish deliberately strip these blades away and proceeds to devour the Sea-urchin with its elastic mouth.

Holmes observes that "the behavior of Echinoderms is certainly complex and plastic to a remarkable degree," but he concludes that the power of forming associations in this group is very doubtful.

FOOTNOTES:

^[7] See Thomson, Prof. J. Arthur, "Secrets of Animal Life," 1919.

SEA-ANEMONES

Some authors find evidences of intelligence in the Cœlenterata, which includes Hydroids, Jelly-fish, Sea-anemones, etc. The Sea-anemone, possessing no nerve-ganglia, when dislodged from its partnership with the Hermit-crab will quickly attach itself to the Crab's leg and again climb up on to the back of the shell, in which the crustacean makes its "home."

Professor J. Arthur Thomson observes that the Sea-anemone "is in some cases more than quiescent in regard to the partnership," and adds that while responsiveness to the touch of the Hermit-crab may have come to be ingrained in its early constitution, "it is difficult to think clearly of its racial establishment."

The behavior of some Cœlenterata, as Holmes remarks, "is often highly plastic and capable of being modified in many ways as the result of previous experience." This is true also of the Vermes ("worms") and Echinodermata. "We do not intend to deny the existence of intelligence in the groups mentioned; we think it not improbable that intelligence of a primitive sort may be discovered, at least in the more highly developed members of these divisions; but at the present time we can only grant the Scotch verdict of 'not proven'."^[8]

FOOTNOTES:

^[8] See, on this phase of the subject, Holmes, "The Evolution of Animal Intelligence," Chapter IX, 1911.

MOLLUSKS

Among the Mollusks, not only the highly organized Cephalopods (Cuttlefish, Octopus, etc.), but even the lowly Oyster, and the more active Snail and Slug appear to possess associative memory.

Prof. A. H. Cooke (Cambridge Natural History, Vol. III), declares that Oysters can learn from experience. Says he:—

“As soon as an Oyster is taken out of the sea, it closes its shells, and keeps them closed until the shock of removal has passed away, or perhaps until the desirability of a fresh supply of water suggests itself. The men take advantage of this to exercise the Oysters, removing them from the sea for longer and longer periods. In time this has the desired effect; the well-educated Mollusk learns that it is hopeless to ‘open’ when out of the water, and so keeps his shell closed and his gills moist, and his general economy in good condition.”

A certain degree of intelligence has been attributed to Snails. Miss Elizabeth Lockwood Thomson, for example, experimented extensively with these Mollusks and found that they are educable—that they can learn by experience.^[9]

FOOTNOTES:

^[9] Miss Thomson’s experiments are described in “Behavior Monographs,” Vol. III, No. 3, 1917, Cambridge, Mass.

OCTOPI

Schneider, Uexküll and Kollman all testify that the Octopus possesses at least a rudimentary intelligence. Romanes agrees with these authorities in attributing to this Mollusk “unmistakable evidences of consciousness and intelligence.”^[10]

“That Loligo (Octopus) and related higher Cephalopods have an extreme agility, resourcefulness, and caution is already fully recognized by naturalists, though abundant observations and experiments are still much needed,” says Prof. John Muirhead Macfarland, in his “The Course and Causes of Evolution” (1917).

A young Octopus in the Naples aquarium was seen by Schneider to attack a Hermit-crab living within a shell upon which were several Sea-anemones. Upon approaching the Crab the Octopus was stung by the Anemones and quickly retreated. Thereafter it avoided contact with the Crab. It had learned by experience that Anemone-protected Hermit-crabs are “good things to let alone.” In the course of time, however, Octopi learn to extract the Crabs from their shells without being stung by the nettling cells of the Anemones. They also learn to capture large Crabs and Lobsters without getting pinched.

FOOTNOTES:

^[10] Romanes, C. J., “Animal Intelligence,” 8th Ed., Page 29, 1904.

CRUSTACEA

Romanes finds in the Arthropoda evidences not only of intelligence, but also (in the Crustacea) of reasoning. Bethe, on the other hand, denies that either associative memory or consciousness exists in any of the Arthropoda (Crabs, Spiders, Insects, etc.) The admittedly complex behavior of these forms is ascribed by Bethe entirely to “reflex action,” wholly devoid of psychic elements.

“This opinion,” says Holmes, “is in part based on *a priori* deductions from the organization of the nervous system and it is held to chiefly by morphologists and physiologists whose observation of the behavior of animals is limited and warped by preconceptions.” Bethe’s experiments were “obviously inadequate.” As Loeb facetiously puts it, “his conclusions are based upon a few spankings he gave a Crab which obdurately rushed into a dark corner (its instinctive action when frightened) despite the presence there of a Devil-fish (Edolene), the Crab’s natural enemy.”

Yerkes and Huggins, Cowles and other observers have found that Crabs and Crayfish are both capable of learning by experience; while Spaulding has demonstrated the capacity of Hermit-crabs (*Pagurus longicarpus*) to form associations.

HERMIT-CRABS

The Hermit-crab deliberately seeks a partner-Anemone if one has not chanced to attach itself to its adopted shell-home. Grasping an Anemone with its claw, it places it upon the back of the Mollusk (e.g., Periwinkle, Whelk) shell which it had appropriated for its residence, and then adds a second and a third Anemone until it is completely “camouflaged.” If the Crustacean has occasion to remove to a new shell (which it does after each moult), it sometimes carries a partner-Anemone on its great claw, as a form of protection. The Anemone is, indeed, a sort of outer defense, being richly endowed with batteries of stinging cells. Thomson remarks that the Hermit-crab’s behavior is “suffused with an appreciative awareness of what he is doing.”

Recent experiments conducted by Mr. S. Mikhailoff, of the Oceanographic Institute at Monaco, on the Hermit-crab proved conclusively that a complex neuro-psychical activity in animals having neither a cerebrum nor a central nervous system is possible. The animal was able to distinguish differences in shades of red, “refusing to respond to anything except the color which was the ‘educating’ stimulus, even when shades of red very near this were employed.” In Man the power to distinguish between colors and their shades is located in the gray matter of the cerebrum. The Hermit-crab has no cerebrum.

An important conclusion derived by Mr. Mikhailoff from his extensive experiments is that it is a mistake to compare the ganglionic nervous system of invertebrate animals to the sympathetic system of vertebrate animals. He also proved by the experiments in question that it is possible to establish an associated reflex “in response to any external stimulus whatever.”^[11]

It is interesting to note in this connection that Crabs, like Birds and other animals, have their own peculiar homing instincts. Carry a shore-crab back from the beach, lay it down, and unerringly it starts back in the right direction, straight for the sea.

“Some land-crabs of the West Indies and North America combine in large swarms to travel to the sea and to deposit therein their spawn; and each such migration implies concert, co-operation and mutual support,” remarks P. A. Kropotkin (“Mutual Aid,” 1902).

FOOTNOTES:

[\[11\]](#) A resume of Mikhailoff’s experiments, translated from the *Revue Général de Science* (Paris), January 15, 1921, may be found in *The Scientific American* for April, 1921. Cf. Morgan, C. L., “Animal Behavior,” 1900.

HORSESHOE CRABS (LIMULUS)

In reference to Horseshoe-crabs Kropotkin says: “I was struck (in 1882, at the Brighton Aquarium) with the extent of mutual assistance which these clumsy animals are capable of bestowing upon a comrade in case of need. One of them had fallen upon its back in a corner of the tank, and its heavy saucepan-like carapace prevented it from returning to its natural position, the more so as there was in the corner an iron bar which rendered the task still more difficult. Its comrades came to the rescue, and for one hour’s time I watched how they endeavored to help their fellow-prisoner. They came two at once, pushed their friend from beneath, and after strenuous efforts succeeded in lifting it upright: but the iron bar would prevent them from achieving the work of rescue, and the Crab would again heavily fall upon its back. After many attempts, one of the helpers would go in the depth of the tank and bring two other Crabs, which would begin with fresh forces the same pushing and lifting of their helpless comrade. We stayed in the Aquarium for more than two hours, and, when leaving, we again came to cast a glance upon the tank: the work of rescue still continued! Since I saw that, I cannot refuse credit to the observation quoted by Dr. Erasmus Darwin—namely, that ‘the common Crab during the moulting season stations as sentinel an unmoulted or hard-shelled individual to prevent marine enemies from injuring moulted individuals in their unprotected state’.”

SPIDERS AND INSECTS

Manifestations of intelligence are much more numerous among Insects and Spiders than in the Crustacea and Mollusca. Forel goes so far as to attribute to Insects an “ability to instinctively draw inferences from analogy.”^[12]

FOOTNOTES:

^[12] Forel, A., “The Senses of Insects,” (Translation) London, 1908.

SPIDERS

Macfarland places the Spiders as following the Cephalopoda in order of intelligence, as does Sir A. E. Shipley. Says the latter:—

“The Arachnida, together with the Crustacea, Insects, Myriapods, and Peripatus, make up the great phylum Arthropoda, a phylum which, from the point of view of numbers of species and individuals, is the dominant one on this planet, and from the point of view of intelligence and power of co-operating in the formation of social communities is surpassed but by the Vertebrata.”^[13]

Spiders possess considerable skill as weavers, hunters, miners, builders and aeronauts. Lacking ears, they “feel” sounds. With but limited visual powers, they nevertheless unerringly pick out the particular strand of the web in which a victim Insect has been trapped. It has been said that the Spider must run to the center of the web before she can know which strand to follow out and reach the entrapped victim. Says Gustav Kafka, in his “Animal Psychology,” “Spiders seem to be endowed with a very keen sense of touch, and know instantly along which strand of their web to go in order to get to any Insect or other object that may become caught in their trap.” The layman probably will wonder why the Spider does not take the shortest path to get its quarry, instead of first going to the center of the web, which involves covering a double distance, perhaps on the very next strand of silk. This is probably due to the Spider’s very limited eyesight, in spite of the six to eight eyes which it possesses—so limited, in fact, that it can see ordinarily for a distance of only a few inches.

Nature has provided the spider with an unusually sharp sense of touch, and it can tell, by the vibration of the connecting strand of silk, that some object has lighted on the web. It speeds to the center of the web, from which, by means of its eight legs and this keen sense of touch it can immediately ascertain by the vibration on which radiating strand the object may have alighted. “That is why all Spiders go first to the center of their web to catch a captive that may lie on the outer radius of the web, and that they may even have passed on the way to the center.”

One cannot but admire, says Prof. E. L. Bouvier, “the marvelous vertical orb-web made by those gifted Spiders which are called indifferently *Acaneus* or *Epeira*. Radii spaced at equal distances form the framework and connect it with the helix where Insects are captured and which adheres tenaciously to the radii. From the center where the latter converge there starts a guide-line which connects them with the ordinary retreat of the animal. Lurking at its post with one hand, so to speak, upon the guide-line, the Spider perceives the slightest tremor of the web. Has an Insect been caught in the net? If so, it runs down the guide-line until it reaches it, binds it fast, and slowly sucks its sweet juices, on the spot if it is small, but on its retreat if it is strong.”

The sense of touch is developed in Spiders beyond all comparison with other animals. “A female orb-weaver, at the center of her web, can tell friend from foe, male from female of her species, an Insect suitable for food from one not suitable, an Insect of a certain size from an inanimate object of the same size, and she can also distinguish between sizes of any two objects which happen to fly or be thrown into her web. This is all accomplished by touch vibrations passing along the radii of the orb on which the eyes of the female Spider rest. Moreover, during courtship of Spiders this system of touch vibrations is utilized as a means of signals to inform the male concerning the proper mood of the female for mating—but pity the dwarfed male should he misinterpret her signals, for instantly she pounces upon him and devours him without showing the least mercy.”^[14]

It has been demonstrated that these little creatures distinguish colors and select a special shade of colors for a background of the marvelously constructed webs. If the observer changes the colors surrounding the web, the Spider inhabitant at once seeks a new location and builds a new web. Spiders must have a sense of light and shade, in order to conceal their webs as they do, and in the case of the so-called “trap-door” variety, to camouflage the cleverly constructed hinged entrances to their tunnel homes so as to resemble their surroundings.

The olfactory pores of both Spiders and Insects are widely scattered over the body, head and appendages. The more highly developed the Insect, the more they are arranged in groups, “most of the groups being found on the legs, wings and mouth parts. So far only a few olfactory pores have been found on the antennæ, these being present on the bases of the antennæ of

Bees, Grasshoppers, Roaches and Crickets. Briefly described, an olfactory pore is nothing more than a nerve passing through a tiny hole in the 'skin' or chitin of the Insect" (McIndoo, *Loc. cit.*, Page 470). "The so-called gustatory sense in Insects is only a phase of the olfactory sense."

Spiders, according to the latest experiments, are deaf, and only a few are able to make sounds. Most Insects can hear, but the Cicada is said to be deaf, and the female both deaf and dumb. Modern research tends to support Forel's conclusion that Insects cannot "hear" in the sense that we do. He compares this perception in them to that in deaf-mutes who feel the rolling of a carriage at a distance. But nothing final can at present be said on this question. Schon, for instance, has described a structure in the tibiæ of Bees which he regards as an auditory apparatus. Child thinks that he has discovered an auditory organ in the Mosquito. Many of the experimental results obtained "indicate that Insects can hear" (McIndoo).

The orb-weaving Spiders have no peers in the art of weaving. They know how to fasten marvelously regular webs between the branches of trees, how to pass over rivers on bridges of floating threads, and even when still young, they know how to use similar threads to take flight through the air as real aeronauts.

"The most difficult but not the most delicate work in the making of an orb-web," says Prof. E. L. Bouvier, "is the establishment of the suspending cable which stretches between two points at a distance from each other and supports the whole structure. Sometimes the Spider fastens its thread at one of these points and then repairs to the other where it stretches and fastens the cord which has issued from its spinnerets during the course of its journey. But this process is not applicable over all sorts of *terrain* and is even practically impossible when the two points are separated by a stream of water or by any other insurpassable obstacle. In this case the Spider stations itself or suspends itself at one of the points ... and emits a thread which is carried by the wind until it attaches itself at another elevated point. According to Fabre the process may differ somewhat, however: the Spider may suspend herself but soon thereafter reascends by her thread; the latter then forms a loop which is stretched out and fastened by the wind as in the preceding cases. In any case the Spider knows quite well when the attachment has occurred. She then stretches her cable and runs back and forth across it several times in order to multiply the number of threads and

thus render the cable more firm. The next thing is to establish another side to the framework: the Spider suspends herself again, then reascends by means of her thread, follows the cable to the opposite end, and then seeks a suitable point further down where she stretches and fastens the thread emitted in the course of the journey.

“In the same manner, or by simply walking from one point to another, a diagonal thread is established which serves as the first radius of the web. Upon this diagonal line a point is chosen to be the center of the structure; the Spider attaches a second radius at this point and then proceeds to walk to the framework where she fastens the other extremity, after which she returns in the opposite direction to stretch this thread and make of it a definite radius; the excess length is reserved at the center to form a cushion. Now at one side and now at the other, in order to render the structure more stable, the Spider attaches new radii by the aid of those already established. When finished the radii are spaced at equal distances; they vary in number according to the species; Fabre counted twenty-one in the angular *Epeira* and thirty-two in the fasciated *Argiope*.

“Resting upon the cushion the Spider now revolves repeatedly about her own axis, attaching to the radii a central helix whose inter-radiary elements are straight lines. Then she advances a little farther and begins to establish a second similar helix which extends to the framework. This second helix is permanent in the *Nephilæ* and temporary in the *Argiopæ*, the *Epeira* and most other forms. Since it consists of cylindrical threads it is not very suitable for purposes of capture. Consequently as soon as it has carried it to the framework the animal returns along this helix, placing between its spirals a new helix whose elements are composed of threads bearing sticky globules. This helix constitutes a marvelously effective trap. In establishing it the Spider takes for a support and scaffold the auxiliary helix; but as the work proceeds the latter is destroyed except among the *Nephilæ*, in which it is retained to give more solidity to the structure.”^[15]

Professor Bouvier does not credit the *Arachnoidea* with much intelligence, and such as they possess he thinks is probably dominated by a strict automatism. Yet he asks, “Is it possible to ascribe to pure automatism or to mere reflex action so judicious a bit of architectural scaffolding? Undoubtedly the psychology of Spiders offers a vast field for observation and experiment.”

It is clear that the Spider is an amazingly good judge of distance, and “can draw parallel lines or converging lines with the accuracy of a draftsman who uses a drawing board and measuring instruments.” It is also certain that they form memory associations and learn by experience: “many examples show that they display a certain degree of discernment when they establish the bases of their shell or nest. And it is by making use of these faculties that they have been able to display plasticity in their habits and undergo an evolution in their industries.

At the present time they are doubtless quite as capable of evolution as in former times, but this tendency escapes our notice because of the automatism which dominates it. We are particularly struck by the extraordinary rôle played by touch in their automatic manifestations. Spiders nearly always perform their labors at night and it is merely by touch that they are able to recognize whether their cables are sufficiently taut, their radii properly spaced, and the spirals of the helix regularly placed. With their legs and their palps they search for contacts and measure distances and the sensitiveness of the spinnerets reveals to them the moment when their silken thread is properly attached. They seem to work as if blind, being largely guided by simple tactile reflexes” (Bouvier).

Says the same authority: “The Avicularidae and the closely related *Atypus* possess in a very high degree the skill of the miner; they also know how to construct masonry, for before weaving for their retreat an envelope of silk they rough-coat it and make it impermeable by means of a mortar made of earth and saliva.... Fabre has likewise studied the manner in which the *Lycosa* of Narbonne builds the bastion which surrounds the opening of its burrow. It forms it ‘of little pebbles, bits of wood, scraps of dry leaves, etc., the whole dexterously interlaced and cemented with the silk.’ And again it is the *chelicerae* which are employed. Many *Lycosas*, especially among the American species, perform similar labors. McCook reports (1889) that the *Lycosa arenicola* builds a bastion in the form of a chimney with small bits of straw or wood and that ... at the base of this edifice it builds a little wall of grains or quartz. More skillful still, the *Lycosa carolinenses* executes a neat bit of basket work; it curves, interlaces and fastens pine needles, so as to form a sort of bastion in the shape of a bird’s nest upside down.”

In discussing the mystery of the Spider's web-weaving dexterity, Prof. J. Arthur Thomson, in his chapter on "Animal Intelligence," ("The Outline of Science," vol. 2) says:

"To credit animals with reason, which means experimenting with general ideas, is, in all probability, too generous. To try to reduce them to the level of automatic machines is certainly too stingy. The fact is that the behavior of animals is often intelligent, often instinctive, and often a subtle mingling of the two. But it is necessary to attach precise meanings to these terms.

"A young Spider, which never made a web before, may make its masterpiece true to the specific pattern the very first time. It does it without any model to copy, and with no trace of the prentice hand. Sometimes it can make the web in the dark, or in the course of a forenoon. This is instinctive behavior, depending on hereditary prearrangements of nerve-cells and muscle-cells, though never without its psychical aspect—a suffused awareness and a background of endeavor. But apart from theory, the fact of observation is certain that inexperienced animals suddenly blossom out into extraordinary intricacies and niceties of behavior, perfect the very first time, not requiring to be learned. This is instinct."

"With reference to the intelligence of Spiders," remarks Mr. Garrett P. Serviss, "I find among Mr. Belt's records an account of the terrible panics caused by the advance of armies of Ants through the forest, all sorts of Insects fleeing wildly before them. But a Spider sometimes escaped by running out to the end of a branch and suspending itself from a single thread of silk, between the enemies above and the enemies below.

"There is exhibited an extraordinary repugnance by many people against admitting that any living being on this earth has been furnished with anything in the slightest degree resembling the peculiar gifts that assure to our race its immensely superior status. This seems to me a petty jealousy. When we dissect the motives of the human heart do we discover any reason why Man should be the exclusive possessor of sparks of Divine light?"

FOOTNOTES:

[\[13\]](#) Cambridge Natural History, IV (1909).

[\[14\]](#) McIndoo, Dr. N. E., “The Senses of Insects,” Annual Report Smithsonian Institution for 1920, Washington, 1922.

[\[15\]](#) *Scientific American*, February, 1920.

BEETLES

Although the Burying Beetles (*Necrophorus*) live an isolated life, generally speaking, they know how to call for help when it is needed, and their appeals for assistance never go unheeded. As is well known, they must have some decaying organic matter to lay their eggs in, thus providing their larvæ with food. But the food must not decay too rapidly, and in order to slow up the process of decay the Beetles bury the corpses of all kinds of small animals.

Occasionally they find the corpse of a Bird or Mouse, which is too heavy for them to “handle” unaided. They thereupon call on their fellow Beetles for assistance, and from four to ten Beetles respond. Uniting their efforts they transport, if necessary, the corpse to a suitable soft ground, where together they bury it. When Gladitsch attached a dead Bird to a cross made out of two sticks, or suspended a Toad to a stick planted in the soil, “the little Beetles would in the same friendly way combine their intelligence to overcome the artifice of Man.” The same evidence of intelligent mutual aid has been noticed among the Dung-beetles.

WASPS

Not many years ago it was confidently asserted that “Man is the only tool-using animal.” Now we know that not only do members of the Ape family employ sticks for weapons and even for crowbars, and also use various objects for missiles, but even in the Insect world we find a creature, the Solitary Wasp (*Ammophila*), which makes a door of soft earth for its nest and then picks up a small pebble in its mandibles and hammers the edges of the door more securely, just as a man would use a pounding-iron. This phenomenon, observed by the Peckhams in 1898,^[16] has been verified by other investigators.

“Before we could recover from our astonishment at this performance,” wrote these now famous observers, “she had dropped her stone and was bringing more earth, and in a moment we saw her pick up the pebble and again pound the earth into place with it. Once more the whole process was repeated, and then the little creature flew away.”

Professor Thomson came to the conclusion that the Wasp’s use of the pebble for a mallet “is not part of the instinctive routine but is an individual touch, probably with more vivid awareness than is associated with the rest of the agency. The difficulty is to think of the origin of either the routine or the finishing touch without postulating intelligence or at least some appreciation of significance.”

Bouvier points out that the use of the little stone is not yet a fixed habit with *Ammophila urania*, belonging “only to certain individuals more highly endowed than others, and is perhaps only accidental even with them. Maybe it will finally pass into the instinctive habits of the species; for the present it belongs to the domain of individual intelligent acts.”

It should here be noted that in the psychological history of the Articulata we witness the gradual transformation of intelligent acts into instinctive acts, whereas the path of evolution in the Vertebrates seems to lead from instinct toward intelligence. It is reasonable to infer, however, that the instincts of the Vertebrates were preceded by an intelligent process and the

establishment of new habits, “which by heredity became part of the patrimony of instinct, modifying the latter and constituting elements essential to its evolution.” (Bouvier, E. L., “La vie psychique des Insectes.”) “It is intelligence,” continues this same high authority, that “regulates by appropriate selection all manifestations of race memory; intelligence again, in the sundry forms of association and individual memory, that puts together the most complicated mechanisms of instincts.”

We are still told by many writers that the Insects are mere reflex machines, despite abundant evidences to the contrary. “This machine,” says M. de Molostwoff, “has no capacity for reasoning and lives as its body directs it to live. Man, however, is the only animal endowed with a will to live according to his reason, [does he?] and he alone is amenable for his actions to his Creator.”

The Wasp may, indeed, “live as its body directs it to live,” but this, according to many modern thinkers, is precisely what Man himself does, his “capacity for reasoning” being conditioned by heredity, largely controlled by “instinct,” and by his bodily structure and internal reactions—actions, reactions and interactions of the vegetative organism (viscera, glands, etc.) as well as the cerebro-spinal system. And just so the *Ammophila* Wasp may be a physico-chemical-biotic machine and yet possess the “capacity for reasoning,” yet at the same time “live as its body directs it to live”—its reasoning power being part and parcel of its organism as a whole.^[17]

Lacordaire, in his well known “*Introduction a l’Entomologie*,” says of Insects:

“If all the instinctive acts of Insects bore constantly the evident imprint of a blind necessity, there would be much less to admire in them than one commonly does. What particularly excites our surprise is that frequently they accommodate themselves to circumstances, and that their acts take on then such an appearance of reason, that it is necessary to look at them closely not to attribute them to a veritable combination of ideas.”

“Insects are largely creatures of instinct,” remarks Professor Thomson, “with inborn capacities for doing apparently clever things, but yet with some degree of intelligence. In an animal’s behavior there is often, no doubt, a mingling of different kinds of activities unified in a way that

baffles analysis. In many cases their behavior under new conditions, their powers of effectively meeting new ends, go beyond mere instinct.”

Prof. J. Lloyd Morgan was led to the conclusion that we have in the case of the tool-using wasp “intelligent behavior rising to a level to which some would apply the term rational. For the act may be held to afford evidence of the perception of the relation of the means employed to an end to be attained, and some general conception of purpose.” Professor Holmes expresses a doubt on this question: “Does she really perceive the relation of means to end? I am not so sure that she does.”

While it is certainly better to be over-cautious than to be inclined to anthropomorphic interpretations where physico-chemical and mechanical or other solutions meet the requirements of the case, there is, nevertheless, such a thing as avoiding an interpretation involving the recognition of a reasoning process in animals just for the sake of “conservatism.” There is no more merit in referring all phenomena to “instinctive behavior” in animals than there would be in attributing all acts of Man to “reasoning.” As matters stand today, the pendulum of thought has swung so far away from the “anthropomorphism of earlier writers” that we are apt to throw a sort of intellectual smoke-screen over a truly rational act by recourse to “the animal’s instinctive actions.”

Again, our “experiments” with animals are by no means a safe guide to accurate estimates of their mental attainments. Measurement of an animal’s real mental endowment may be far more safely made by observation of its behavior in its natural environment. That this is true is fully recognized by scientists who deplore the tendency of some observers to draw conclusions before becoming thoroughly acquainted with the general behavior of the forms observed. “On the other hand,” says Professor Holmes, “one is tortured by the feeling that our experimental methods often fail to give us a true measure of an animal’s possible attainments, and that it is just in meeting exceptional situations which occur in the animal’s natural course of life that the highest manifestations of its intelligence are reached.”

Wasps of the genus *Eumenes* are said to mould tempered earth into pottery of the most artistic design.

Fabre tells us that the ringed *Calicurgus* Wasp first stings its captured Spider in a spot near the mouth, paralyzing the poison claws. This

precaution being taken, it proceeds to pick out the thinnest part of the Spider's armor, between the fourth pair of legs, driving in its poison needle with a skilled surgeon's precision.

The Two-banded *Scolia* Wasp lays up as food for its larva a Rose-chafer grub. A single grub is the sole provender for the larva for the several weeks from its hatching until its entrance into the cocoon stage. It has frequently been stated that the "head of game" must remain fresh all the time the Wasp larva is feeding on it, or the Wasp larva dies. To remain fresh it must stay alive until the larva takes its last mouthful.

To insure this, a complicated course must be followed with delicate precision by both the mother Wasp and the larva. With a single and seldom-erring stroke of its sting *Scolia* paralyzes the nerve-centers which control motion on the Rose-chafer grub—a stroke which must reach a buried nerve-center no more than a fiftieth of an inch across, and which must at the same time miss the nerve-center which keeps life in the victim.

The infant *Scolia* must then, to keep its food fresh to the end, so eat its living but paralyzed grub that the vital parts are left to the last meal. Accordingly, the *Scolia* lays its egg always at the precise spot on the Rose-chafer grub where the Wasp larva must take its first mouthful. There the larva inserts its head and never withdraws it until the grub is completely devoured. Fabre found by repeated experiment that if he disturbed the larva the chances were even that it lost the clue to its selective meal, killed its "game" and died of ptomaines. If he moved the recently hatched larva to another starting-place on its food supply it was infallibly lost. So it was also when Fabre gave it as food another grub, even though closely related to the Rose-chafer. It tried to eat the strange nutrient, but somehow never succeeded. Hence, according to most authorities at least, it is a prerequisite to the survival of *Scolia* that the mother Wasp select only the Rose-chafer grub, capable of being paralyzed in such a way as to leave the grub alive but without power to move. In all, there are four critical conditions to be met in order that the progeny survive: and they are met.

In their studies of the Mason Wasp, *Odynerus parietum*, the Peckhams found that, contrary to Eimer, the grubs stored for food for the larvæ were by no means all paralyzed, and that in most nests several caterpillars died.

In some cases all the grubs died, yet the Wasp larvæ fed upon them without apparent injury or dissatisfaction.

Holmes remarks that while we may not be compelled to admit that Wasps have “ideas,” it must be granted, he thinks, “that a Wasp which after cutting a caterpillar in two and carrying away one part, came back and searched diligently for the remainder” retained, somehow, “an impression of the missing part and its location. If out of sight it was not out of mind.... If the Wasp does not have an idea of its prey it has something which plays a rôle similar to that of ideas in ourselves.... If there is something representing ‘part-of-caterpillar-among-leaves’ that leads the Wasp on its hunt, we may conclude that there is something corresponding to ‘part-of-caterpillar-now-in-nest’ which prevents further search.”

FOOTNOTES:

[16] Peckham, G. W. and E. G., “Wasps, Social and Solitary,” Boston, 1905.

[17] Cf. Loeb, Jacques, “The Organism as a Whole,” New York, 1916, and “The Dynamics of Living Matter,” New York, 1906; and Berman, Louis, “The Glands Regulating Personality,” New York, 1921.

BEES

Sir John Lubbock long ago declared that “if we judge animals by their intelligence as evidenced in their actions, it is not the Gorilla and the Chimpanzee, but the Bee, and above all the Ant, which approach nearest to Man.”

While Bees, Ants and Termites indisputably possess some degree of intelligence, as distinct from what we are pleased to call “instinct,” it is doubtful if any naturalist or comparative psychologist of today would agree with the eminent British scientist in this generalization. Yet Man has, perhaps, more to learn from the example of Bee life than from the more intelligent activities of the simian world.

Prof. W. F. Wilson, of the University of Wisconsin, in a recent lecture, remarked that a Bee knows when it has had enough, and is satisfied to work in a self-selected domain before it “grabs for more.” For instance, if a Bee settles itself to work in one corner of a field, it will not leave that corner until it has exhausted the nectar in all flowers in that corner. It will not fly to another corner after having found one flower deficient. Neither will it quit one plant until it has tried its tongue in all the flowers of that plant. This has been proved by releasing Bees with different colored powder spread on them and then watching them at work.

Professor Wilson stated that another experiment demonstrated that Bees have some mathematical ability. In this experiment it was found that the Bees realized when a sufficient number were working on one apple tree and no more tried to get on it.

Another peculiarity noticed by Professor Wilson is this: While Bees are at work they will not change from a flower of one color to one of another color until the nectar has been exhausted in the flowers of the first color. That is, if the Bee begins work on a blue aster, it will not go next to a red one, but always seeks a blue one again. This fact was noted by Aristotle in the fourth century B. C., though the belief is still widely prevalent that Bees “fly about from flower to flower in a haphazard way.” As a rule “Bees keep to a single species of flower for collecting pollen and nectar.”

Nevertheless, Bees are partially color-blind, as is evidenced by the experiments of two German biologists, Prof. F. Frisch and Lothar Tiralá. “These investigators have shown that to the Bee, red and black look alike, orange and yellow look the same as green, and that there is no difference in the appearance of blue, violet and purple. But Bees have one advantage over Man; they can see the rays of ultraviolet light, which are invisible to our eyes.

“It was also discovered that the mysterious guiding influence by which the Bee is brought back to its hive is nothing more than experience. It has long been known that Bees find their way home sooner the longer they have lived in their hive. To test this common-sense view Bees were put to sleep by ether, taken to a new hive, and moved some twelve yards away. None of them could find their way back to the hive until the third day afterward, when 30 per cent got home. By the eighth day, however, 90 per cent of them had learned to find their way back to the hive.”^[18]

Some experiments by Professor Young, of Geneva, antedating those of Professor Frisch, also showed that Bees “build up a knowledge of the country about the hive.” On the other hand, it seems to be well established that Bees, like most animals, have a true homing sense, or sense of direction. “Even Bees with their eyes obscured have been known to make a ‘bee-line’ for the hive from considerable distances,” says Thomson.

When, by exception, Bees build their nest in the open, they invent arrangements which are new and in the true sense intelligent to meet the new conditions (Bergson). In the face of cumulative evidence, the great Fabre was forced to modify his theory of immutable instinct, and grant to Insects a modicum of “discernment,” since they have the capacity of learning by experience.

Professor Frisch published in the Munich *Medizin Wochenschrift* some observations upon the means of communication employed by Bees. “He placed a dish of sugar solution on a table by an open window. Shortly after a chance Bee had noted this and flown off with booty therefrom, the dish was crowded with Bees. When it was removed they quickly disappeared, save for an occasional reconnoiterer. When a fresh dish was set out they quickly reappeared in quantities. By touching the back of each Bee with a

spot of color, the experimenter then perceived that subsequent Bees had been sent, and not escorted.

“The conduct of the rediscoverer on her return to the hive was next noted. She first gave over her plunder to the workers, and then executed a curious dance, describing circles and other figures. Her audience watched her attentively and attempted to touch her. When one of the marked Bees succeeded in doing this, the latter at once made her exit and flew to the feeding place; but the unmarked Bees soon ceased to pay her any attention. It appears that there is here some means of communication based upon touch rather than upon sight or hearing; and that it is adequate for giving information as to the presence or absence of food, but inadequate to give its location unless it be already known to the recipient of the message.

“Experiments with two dishes of food at a considerable distance apart verified this. As before, after they had once been discovered the dishes were removed and ultimately replaced; but when replaced, the ‘white’ dish only was filled, the ‘yellow’ one being left empty. The ‘white’ dish was rediscovered by a ‘white’ Bee; and when the latter returned to the hive not only the ‘white’ but also the ‘yellow’ Bees responded to her dance, left the hive and flew to their respective dishes, the ‘yellow’ Bees of course having the search in vain. As before, unmarked Bees ignored the dancer.

“That there is a little more flexibility to the signal system than this might indicate appeared when natural conditions were imitated, linden and acacia blossoms being offered respectively to groups of Bees accustomed to seek these. The dancing linden Bee now occasioned excitement only among the linden Bees, and not among the acacia group. The same distinction was made when two dishes of sugar were differently perfumed, suggesting that scent rather than actual modification of the signals may have been responsible. When blotting paper saturated with sugar-water was used instead of the dishes, the Bees found some difficulty in sucking the fluid up, and returned only half laden. They did not then trouble to perform the dance, showing that this is reserved for exceptionally rich finds.”^[19]

The Bee is good-natured and even long-suffering, but there are limits to its patience or generosity. So long as nectar is superabundant, the Bee allows the drones to live in the communal hive without rendering any service in return. “But one day the decree goes forth that those who do not

work shall not eat, indeed shall not live.... Vigorously and pitilessly the long-suffering workers at last turn on the drones and slay them all.”^[20]

FOOTNOTES:

^[18] *Scientific American*, April, 1924.

^[19] *Scientific American*, April, 1922.

^[20] “*The Outline of Science*,” Vol. II, edited by J. Arthur Thomson, 1922.

ANTS

Many scientists have asserted that, next to Man, Ants have the most intelligence of any living creature. Charles Darwin said “the Ant’s brain is one of the most marvelous atoms of matter in the world, perhaps more so than the brain of Man.”

In the opinion of the present writer the anthropoid apes stand next to man in intelligence. But it is quite true that in their social organization and industrial activities Ant “civilization,” in some respects, approximates human “civilization.”

Ants plant and harvest crops, domesticate animals, have a social system which includes working slaves and a military caste, a police force and jails. They perform astonishing feats of engineering. They have efficiency, initiative, and productivity without profiteering. Cooperation, individualism and “patriotism” exist side by side in a real *commonwealth*. There is a division of labor, but no exploitation of the many for the benefit of the few. All classes share equally in the benefits of their common toil, if we accept the highly specialized warrior ants as rendering military service equivalent to the work of the so-called “slaves.”

Members of the military caste do not work, and are fed by the slave population. But this is due to high specialization rather than to imposition or snobbery. The erstwhile working mandibles of the soldier have gradually become transformed into veritable sabres or bayonets. They could obtain their own food if they so desired, but they have become absolutely dependent on the “working class” for their daily bread.^[21] But in return they protect the colony, even at the sacrifice of their own lives.

Each worker Ant finds its own task, and willingly contributes its share. There are no slave-drivers. When one shift of workers has become fatigued, or must stop to partake of food, its place is taken by another shift of equally skillful workers. No time is lost.

If one of the workers becomes encumbered with dirt, its mates come at once to its assistance with “first aid,” cleaning the unfortunate one by

brushing and washing. During their mining operations in digging holes and removing stones, an Ant is often injured, whereupon others rush to its aid and carry it to a quieter gallery, where it is by no means neglected.

Above all animals of this planet, Ants know the value of mutual aid, though co-operation is practised among Insects of many kinds.^[22]

Forel points out how it is a common practice among many species of Ants for one which possesses an abundance of food in its crop to share it with any of its less fortunate comrades who may apply for it—that is, with any member of the same nest or colony of nests. Approaching each other, they exchange a few signals—movements—of the antennæ, and, says Sir John Lubbock, “if one of them is hungry or thirsty ... it immediately asks for food.” The well-supplied Ant sets apart its mandibles, takes the appropriate position, and regurgitates a drop of half-digested food—a transparent fluid—which is licked up by the hungry Ant. Forel concluded that Ants possess a divided digestive tube, the posterior part being for the special use of the individual, the other, the anterior part, being used chiefly for the benefit of members of the commune. Any Ant which, possessing a crop full of food, refuses to feed a needy comrade, is treated as a “criminal” or outcast. During “war times” such an Ant is treated as a “traitor,” and attacked by its kinsfolk with greater fury than is exhibited toward the foreign enemies of the species. On the other hand, if an Ant has fed an Ant belonging to the enemy species, it will be treated by the kinsfolk of the latter as a friend.^[23]

It is well known that the most convincing evidences of intelligence are to be found among the social Insects, where mutual aid is “the order of” every day.

When the able naturalist, Thomas Belt, was superintending a gold mine in Nicaragua, he kept close watch on the animal life about him, large and small. He tells us that he once saw a wide column of Ants attempting to pass along a crumbling, nearly perpendicular slope.

“They would have got very slowly over it, but a number having secured their hold, and reaching to each other, remained stationary, and over them the main column passed.

“Another time they were crossing a watercourse along a small twig, not thicker than a goose-quill. They widened this natural bridge to three times its width by a number of Ants clinging to it and to each other on each side, over which the column passed three or four deep. Except for this expedient they would have had to pass over in single file, and treble the time would have been consumed.

“Can it not be contended that such Insects are able to determine by reasoning powers which is the best way of doing a thing, and that their actions are guided by thought and reflection?”

In 1921, it was discovered that the interior woodwork of the Livestock Exchange Building in Wichita, Kansas, was being tunneled through by Ants. Flypaper was placed across their line of march. The Ants thereupon made sufficient sawdust to cover the sticky flypaper and went on with their “industry.”

Ants domesticate animals, for the same reason that Man does. Among other animals thus domesticated are herds of Plant-lice, or “green-flies,” which are to them the equivalent of our dairy animals. The Ants’ “cow” (Aphid) secretes a “honey-dew.” In order to increase the supply of the sweet excretion the Ants gently stroke the Aphids. When the sap supply for the Aphids fails, the Ants carry their “cows” to new food plants, and when winter comes on both the adult Plant-lice and the eggs are carried out of reach of frost into the Ant caverns and carefully attended until spring, when they are again placed on the swelling plant-buds.

The presence of certain little Crickets, Beetles, and a certain species of Caterpillar in the nests of the Termites and the true Ants, who seem to enjoy their presence, has recently been explained on the ground that the Ants like the odor of these strange “guests.” It has been noted that the Beetles give off more “fragrance” upon being caressed by the Ants. “One species of Ants carries Mites about on the body, feeding them and caring for them, but apparently deriving no benefit from them. Evidently Ants are fond of keeping pets!” (Thomson).

Professor Thomson (in the “Outline of Science,” Vol. II) writes delightedly of the varied activities of Insects, and concerning certain species he says:

“The Tailor-ants, common in warm countries, make a shelter by drawing leaves together, and their co-operative hauling is admirable; their mandibles are their needles, if you like, but they have nothing to sew with: what does each do but take a larva in its mouth so that the silk secreted from the offspring serves as thread for the parents?”

Again: “A common Harvesting-ant of South Europe collects seeds of clover-like plants, lets them begin to sprout so that the tough envelopes are burst, exposes them in the sun so that the germination does not go too far, takes them back underground and chews them into dough, and finally makes this into little biscuits which are dried in the sun and stored for winter use. Many ‘White Ants’ or Termites grow mushrooms in extensive, specially constructed beds of chewed wood, and some of the true Ants show a similar habit.”

That Ants have some means of communicating one with another goes without saying, but our knowledge of Ant language is still very meager, despite the long and patient labors of many myrmecologists. Some of the actions considered to be involved in communication are striking with the antennæ, butting with the head, opening the jaws, beating the ground with the abdomen, and the production of sounds by various kinds of apparatus for stridulation.

“While Ants may not be able to talk about things in their sign language,” says Professor Holmes, “they apparently express their different feelings and inclinations in ways which are intelligible to other Ants. Wasmann has compiled a sort of vocabulary of signs made by the antennæ—a ‘Wörterbuch der Fühlersprache,’ which is about as extensive as Mr. Garner’s languages of Apes. According to the vigor and frequency of the strokes of the antennæ, and the part of the body stroked, the Ant which is addressed may be importuned for food, warned of danger, or induced to co-operate with the communicants in various activities.”

FOOTNOTES:

[21] Cf. Lubbock, Sir John, “Ants, Bees and Wasps,” 1883; Wasmann, E., “Comparative Studies in the Psychology of Ants and of Higher Animals,” 1905; Forel, A., “*Recherches sur les fourmis de la Suisse*,” Zurich, 1874; “The Senses or Insects,” 1908. See also Beebe, William, “The Edge of the Jungle,” 1921.

[22] The value of mutual aid as a factor in evolution was dimly divined by Goethe, and was first expressed as a “law” by Professor Kessler in 1880, who was then Dean of the St. Petersburg (Leningrad) University. Having read Kessler’s lecture in 1883, Prince Kropotkin began a series of articles on the same subject, resulting in the publication in 1902 of his great work, “Mutual Aid as a Law of Nature and a Factor of Evolution.”

[23] Cf. Huber, Pierre, “*Les fourmis indigènes*,” Geneva, 1861, and Forel, loc. cit.

TERMITES (“WHITE ANTS”)

The organized social life of the “White Ants” or Termites is of unusual interest for the student of animal intelligence. The Termites have kings, queens, soldiers and workers.

Not related to the true Ants, the Termites are not unlike our Roaches in the construction of their bodies. Though found in the United States and in Europe, their main habitats are Africa and Australia. In the latter countries their so-called nests are of prodigious dimensions, exceeding sometimes five hundred times the length of the Insect (ten to twelve millimeters), thus surpassing the tallest buildings constructed by Man relative to the height of the builders. While the height of Termite nests are sometimes more than 500 times the length of the Insect, the Eiffel Tower is but 175 times as tall as the workers who reared it. “The number of dwellers in these nests exceeds by count the number of inhabitants of many large countries” (Molostwoff).

CONCLUSION

In view of the foregoing facts illustrative of the mental life of Invertebrates, there remains little ground for denying to creatures below the backbone phyla a certain modicum of intelligence, well in accord with their needs. That any Invertebrate is capable of the process of reasoning is, of course, a debatable question, but the existence of mental processes in these groups is, apparently, indisputable, and where there is *mind* there is *intelligence*, however limited it may be.

Forel attributes to Insects passions closely akin to those exhibited by the higher Vertebrates, though these vary considerably with the diverse species. Wasps, certain species of Ants, and a few Beetles, are extremely irritable and pugnacious. Among the less intelligent species no passions are manifest apart from hunger, thirst and sexual appetite. The memory likewise varies according to the species, and, as might be expected, is at a minimum in the small-brained forms and most highly developed in the social Hymenoptera (Bees, Ants and Wasps).

“It must be admitted,” declares Forel, “that Insects are capable of perceiving, of learning, of recollecting, of associating their recollections and of utilizing them to accomplish their ends. They have various emotions and their will is not purely instinctive, but offers individual plastic modifications, adapted to circumstances.”

Although many authorities are inclined to think that Forel goes too far in his estimates of Insect Intelligence, Bouvier’s^[24] extensive studies have led him to practically the same conclusions. He rejects Bethe’s claim that Insects are mere reflex machines, because they can adapt themselves to circumstances, acquire new habits, learn to remember, and manifest discernment.

Comparing their organization with ours, McIndoo concludes that Insects “have perhaps accomplished more than we have.... Furthermore some Insects, for example Honey Bees and Plant-lice, have evolved methods for controlling sex; this subject has probably puzzled Man as much as life

itself, yet Man can neither control sex nor knows how to control it.... Let us cease looking with scorn upon Insects.”

“We prize so highly all our own aptitudes as to believe that they are unequalled, even when inspired by the least commendable motives,” says Bouvier. “Though bellicose ourselves, we think it strange that beehives or ants-nests should engage in warfare. At times we revert to barbarism by reducing our enemies to slavery, yet we exclaim with surprise at the habits of slave-making Ants.”

There is nowhere any distinct break in the evolutionary series—no fundamental distinction between the animal and the human mind. Protoplasm is protoplasm, wherever found, and mind is mind wherever it becomes manifest. There can no more be two totally distinct and fundamentally different kinds of mind than there can be two or more totally distinct kinds of protoplasm, one human, the other sub-human. The Amœba and Man are both the product of protoplasmic differentiation, and the primordial protoplasmic cell embodied in its substance all potentialities of Life and Mind upon this planet.

FOOTNOTES:

[24] Bouvier, E. L., “*La Vie Psychique des Insectes*,” 1918.

Transcriber's Note:

The chapter "Horseshoe Crabs (*Limulus*)" was missing from the original table of contents, so it was re-added.

Footnotes were moved to the ends of the chapters in which they appeared.

Minor punctuation errors have been changed without notice.

Spelling was retained as in the original except for the following changes:

Page 13 : "we come to the Amoeba"	"we come to the Amœba"
Page 19 : "highly organized Cephelapods"	"highly organized Cephalopods"
Page 27 : "its post with o e"	"its post with one"
Page 29 : "through a tiny hold"	"through a tiny hole"
Page 31 : "establishing it the Spide"	"establishing it the Spider"
Page 34 : "remarks Mr. Garret P. Serviss"	"remarks Mr. Garrett P. Serviss"
Page 41 : "The Two-banded Scolla"	"The Two-banded Scolia"

*** END OF THE PROJECT GUTENBERG EBOOK THE
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